Team members:

For each of the following contexts I will ask you to identify the components we learned in class. Note, sometimes when I ask you for a value, that value is actually unknown.

Example Question

Of all freshman at a large college, 16% made the dean's list in the current year. As part of a class project, students randomly sample 40 students and check if those students made the list.

(a) What could be the research question?

(b) What is the population of interest?

(c) Describe the population parameter we want to infer. Also, what is its value?

(d) Describe the sample statistic we can use to infer the population parameter. Also, what is its value?

(e) Describe the sample we have. Would you expect this sample to be representative of the population of interest? Why?

Question 1

In a random sample 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt.

(a) What could be the research question?

(b) What is the population of interest?

(c) Describe the population parameter we want to infer. Also, what is its value?

(d) Describe the sample statistic we can use to infer the population parameter. Also, what is its value?

(e) Describe the sample we have. Would you expect this sample to be representative of the population of interest? Why?

Question 2

As part of a quality control process for computer chips, an engineer at a factory randomly samples 212 chips during a week of production to test the current rate of chips with severe defects. She finds that 27 of the chips are defective.

(a) What could be the research question?

(b) What is the population of interest?

(c) Describe the population parameter we want to infer. Also, what is its value?

(d) Describe the sample statistic we can use to infer the population parameter. Also, what is its value?

(e) Describe the sample we have. Would you expect this sample to be representative of the population of interest? Why?

Question 3

A nonprofit wants to understand the fraction of households that have elevated levels of lead in their drinking water. They expect at least 5% of homes will have elevated levels of lead, but not more than about 30%. They randomly sample 800 homes and work with the owners to retrieve water samples, and they compute the fraction of these homes with elevated lead levels.

(a) What could be the research question?

(b) What is the population of interest?

(c) Describe the population parameter we want to infer. Also, what is its value?

(d) Describe the sample statistic we can use to infer the population parameter. Also, what is its value?

(e) Describe the sample we have. Would you expect this sample to be representative of the population of interest? Why?